

Flood risk assessment at the neighbourhood level using Spatially Adaptive Weighting

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Abstract

Floods pose a significant risk to urban areas, however assessing the magnitude of the risk objectively to population is challenging. Here, this study conceptualises this problem and develops a spatially adaptive flood risk framework for 412 dense urban neighbourhoods along the Maribyrnong River catchment in Melbourne, Australia. Multiscale Geographically Weighted Regression (MGWR) was used to derive objective, spatially varying indicator weights. Indicators were classified and normalised under hazard, exposure and vulnerability components following the IPCC risk framework. Flood depth and spread was simulated from a selected extreme flood event using high-resolution modelling while multiple socioeconomic neighbourhood profile variables were selected to compute exposure and vulnerability indicators. Results show that vulnerability and exposure exert greater influence on flood risk than hazard alone, with population density, building density and vegetation identified as the dominant risk drivers. The highest composite flood risk was identified in the area with the highest flood exposure. Contrastingly, neighbourhoods south and south-west emerged as previously unidentified high-risk areas despite their distance from the primary waterway and floods. The framework demonstrates that a spatially explicit, objective weighting approach captures neighbourhood-level disparities. This offers a replicable methodology for flood risk assessment and resilience planning in urbanised catchments to support policy insights.

1. Introduction

Floods are an intensifying frequent natural hazard which causes long-lasting economic impacts of over \$1 trillion in global building damages since 1980 (Rathnasiri et al. 2023). Urban areas experience flood events at a higher intensity as the land use and land cover (LULC) influences the flood water's runoff and accumulation where urbanisation increases the number of people and property exposed (Ajtai et al. 2023; Miranda et al. 2023). As climate change and urbanisation continue to intensify flood impacts, robust methods for assessing flood risk are becoming increasingly important for supporting local planning and disaster risk reduction. Australia has experienced increased rainfall from the 2020–2023 La Niña event (Huang et al. 2024). Consequently, this increases the intensity and frequency of flooding where this was most evident in the east and southeastern areas of Australia, impacting areas with a flooding history.

Following the IPCC (Intergovernmental Panel on Climate Change) conceptual framework on climate risk assessment, flood risk can be considered as the interaction between hazard, exposure, and vulnerability (IPCC 2023). Hazard represents the flood event and the physical processes that influence its occurrence. Exposure refers to the people, property, and environmental assets that may be affected, while vulnerability describes the environmental, physical, and socioeconomic conditions that influence the severity of flood impacts. The IPCC framework is adopted in this study to provide a consistent classification of indicators, addressing the inconsistencies found in previous studies where similar indicators are assigned to different risk components.

A major source of conceptual inconsistency in the flood risk literature is the absence of a consistent basis for allocating indicators across risk components. For example, elevation may be classified as a hazard

indicator in studies examining flood propagation, but as a vulnerability indicator in studies concerned with the susceptibility of low-lying areas to inundation (Feng et al. 2023). As a result, the same variable may be categorised differently across studies, limiting comparability and reducing the clarity of indicator contributions to overall flood risk. This consequently impacts the magnitude of each risk component index across studies depending on how many indicators are sorted under each component and whether all three components were used in the assessment. For example, a study using hazard and vulnerability may overestimate the vulnerability of a neighbourhood compared to an identical study that uses hazard, vulnerability and exposure. Frameworks that incorporate all three risk components simultaneously (Armenakis et al. 2017; Feng et al. 2023; Huu Duy et al. 2024) may not follow a consistent definition of hazard, vulnerability and exposure which presents another conceptual gap. For example, Armenakis et al. (2017) uses exposure as another form of vulnerability while hazard includes areas susceptible of flooding rather than solely being the flood waters themselves. Huu Duy et al. (2024) uses Land Use Land Cover (LULC) as an exposure indicator while Feng et al. (2023) specifies between different LULC types before allocating them to exposure and vulnerability individually. Although Feng et al. (2023) follows the IPCC definitions and considers the different LULC types, the inconsistency of indicator and risk component allocation remains a prevalent occurrence in the literature.

Despite substantial advances in flood risk assessment, two methodological gaps remain in indicator-based approaches: the derivation of objective indicator weights and the representation of spatially varying indicator contributions to flood risk. The Analytical Hierarchy Process (AHP) is the most prevalent weighting method for flood risk assessments (Berrezel et al. 2026) but its subjective nature makes it liable to biases. Objective weighting methods used for flood analysis have been Frequency Ratio (FR) for susceptibility assessment (Yilmaz 2022) and PCA for social vulnerability assessment (Ajtai et al. 2023), although neither were used for risk. The Frequency Ratio performed better than AHP but AHP-FR returned better results (Yilmaz 2022). A more advanced method proposed by Chen (2021) using conceptualised PCA-GWR method returns index values that ranks flood risk using integers from 0 to 8. However, this index ranking introduces the loss of data granularity which limits the assessment if neighbourhoods received similar values. Although these advances provide methods without subjective judgement, indicator-based flood risk assessments typically rely on spatially invariant indicator weights, limiting their capacity to represent the heterogeneous drivers of flood risk across space. Nevertheless, the spatial variation in indicator contributions remains largely unaddressed in existing indicator-based flood risk assessments, despite its importance for capturing local differences in flood risk. Furthermore, existing flood risk weighting methods require a trade-off between accurate results at the cost of biased judgement or objective methods with moderate to poor accuracy. Another issue is that all the weighting methods are applied to the entire study area rather than accounting for the spatial variation across the study area. Geographic and urban factors such as LULC and population exhibit significant variations across larger study areas when analysed at a local level.

In this context, there is a need for a flood risk assessment framework that moves beyond globally applied weighting schemes by integrating objective indicator weighting with spatially varying relationships among risk components. This study addresses these challenges by developing a spatially explicit flood risk

assessment framework that combines the IPCC risk components with local modelling to derive neighbourhood-specific indicator weights without relying on subjective judgement. The framework quantifies the contribution of hazard, exposure, and vulnerability indicators to flood risk while accounting for spatial heterogeneity across the study area. By integrating objective weighting with geographically varying relationships, the proposed approach provides a more locally representative assessment of flood risk than conventional global weighting methods. The central research question is: *How does flood risk vary across neighbourhoods, and how do socioeconomic, urban, and physical indicators influence its spatial distribution and the relative contribution of hazard, exposure, and vulnerability?*

2. Methodology

2.1 Study Area and Risk Assessment Framework

The study area is the Maribyrnong River catchment in Melbourne, Victoria, Australia, whose neighbourhoods have an extensive flood history, with 28 recorded events since 1871 (Water 2023), and riverine flooding occurs when prolonged heavy rainfall saturates the upper catchment, reducing infiltration capacity and increasing overland flow to the point where the Maribyrnong River overtops its banks and inundates the low-lying floodplain areas downstream, a process accelerated by the progressive replacement of permeable surfaces with impervious urban cover that reduces travel times and increases peak flows (Miranda et al. 2023). The October 2022 event, the third largest on record for the catchment, is used here as the basis for the hazard simulation because it generated the most spatially complete sensor record available and because its magnitude produced inundation across the full extent of the study area rather than confining flood waters to the immediate river corridor, conditions that are necessary for a neighbourhood-scale assessment of the kind developed in this study. Despite this history and the ongoing flood threat the catchment faces, systematic flood risk assessments of the catchment remain limited.

The study area comprises 412 SA1 neighbourhoods distributed across the local government areas of Maribyrnong and Moonee Valley in Melbourne, covering a catchment that drains to the Maribyrnong River and whose urban morphology ranges from dense residential and industrial areas in the south and centre to suburban and peri-urban land covers in the north and west (Fig. 1). The SA1 is the smallest geographic unit for which the Australian Bureau of Statistics publishes census and SEIFA data, and its use here reflects the need to capture intra-urban disparities in exposure and vulnerability that would be obscured at the SA2 or local government area scale, where the averaging of socioeconomic characteristics across larger and more heterogeneous populations conceals the neighbourhood-level conditions that determine the actual experience of flood risk.

The proposed risk assessment framework combines a GIS-based spatial multi-index approach with a hydraulic simulation of a historical flood event, structured across four phases that move from indicator identification through data processing, weighting and index calculation to produce a neighbourhood-scale flood risk index for the 412 SA1 units of the Maribyrnong River catchment (Fig. 2). The sequence of phases is not arbitrary but reflects a logical dependency in which each phase produces the inputs that the next requires. The indicator classifications established in Phase 1 determine the data collected in Phase 2,

the normalised indicator values produced in Phase 2 are the inputs to the weighting procedure of Phase 3, and the neighbourhood-specific weights derived in Phase 3 govern the index calculation of Phase 4.

Two design choices distinguish this framework from the approaches most commonly found in flood risk literature. The first is the use of Multiscale Geographically Weighted Regression (MGWR) to derive indicator weights that vary from one neighbourhood to the next rather than applying a single global weight across the study area, which addresses the limitation of both the Analytical Hierarchy Process, whose weights reflect expert judgement rather than the spatial structure of the data, and equal weighting, which assumes that all indicators contribute equally regardless of the local context in which they operate. The second is the use of the IPCC definitions of hazard, exposure and vulnerability as the basis for indicator classification, which provides a standardised and replicable framework for allocating indicators to components in place of the inconsistent conventions that characterise much of the existing literature (Feng et al. 2023).

2.2 Phase 1: Indicator Identification and Classification

2.2.1 Indicator selection and classification

Fourteen indicators were selected across the three components, covering the physical hydrology of the catchment, its urban morphology and the socioeconomic profile of its resident population, and are presented with their units of measurement, normalisation direction and data sources in Table 1, where the normalisation direction indicates whether higher values of an indicator correspond to higher flood risk, denoted by a positive sign, or whether lower values correspond to higher risk and the normalised value is reversed before component calculation, denoted by a negative sign.

Table 1
Indicators by component, unit of measurement, normalisation direction and data source.

Component	Indicator	Unit	Norm.	Source
Hazard	Mean flood depth	m, mean per SA1	+	HEC-RAS simulation
	Flood spread	% per SA1	+	HEC-RAS simulation
	Drainage density	km ⁻¹ , mean per SA1	+	10 m DEM (DELWP 2025)
Vulnerability	Mean elevation	m, mean per SA1	-	10 m DEM (DELWP 2025)
	Mean slope	°, mean per SA1	-	10 m DEM (DELWP 2025)
	Mean curvature	m ⁻¹ , mean per SA1	-	10 m DEM (DELWP 2025)
	Economic Resources (SEIFA IER)	Raw SEIFA score	-	SEIFA 2021 (ABS 2023)
	Vegetation density	% per SA1	-	20 cm tree extent raster (DELWP 2021)
	Transport density	% per SA1	-	Road casement shapefile (DTP 2022)
Exposure	Population density	People per m ²	+	ABS Census 2021 (ABS 2021a)
	Building density	% per SA1	+	Maribyrnong/Moonee Valley buildings (Council n.d.); Global ML Building Footprints (Microsoft 2023)
	Dependent population	% per SA1	+	ABS AGE10P (ABS 2021a)
	Population with health condition	% per SA1	+	ABS ASSNP (ABS 2021b)
	Education–Occupation Status (SEIFA IEO)	Raw SEIFA score	-	SEIFA 2021 (ABS 2023)
<i>Note: Norm. = normalisation direction. + denotes positive normalisation; – denotes negative normalisation.</i>				

The indicators use socioeconomic and demographic data drawn from Census data and SEIFA release, the physical terrain derivatives from a DEM representing conditions that change gradually over time, and the flood depth and spread from the October 2022 event, a temporal spread that is consistent with data availability constraints in urban catchment studies and that introduces no material inconsistency given

that the physical morphology of the catchment and the broad socioeconomic structure of its neighbourhoods are unlikely to have changed substantially across this period.

Several allocation decisions diverge from conventions found elsewhere in the literature and are grounded in the distinctions of definitions outlined above. The clearest case is vegetation density, which is classified under exposure in studies where it represents an asset requiring protection but is placed here under vulnerability, since the riparian and floodplain vegetation of the Maribyrnong catchment was established specifically to intercept and absorb floodwater and to reduce surface runoff (Morgan 2021). The vegetation's presence reduces the conditions under which flood impact worsens rather than constituting an element at risk. A related distinction applies to the population indicators: dependent population and population with health conditions are placed under exposure rather than vulnerability because both identify sub-groups of the resident population who are present within the hazard zone and directly affected, which corresponds to the IPCC definition of exposure as the presence of people or assets in areas subject to potential harm (IPCC 2023), while the consequences of their reduced mobility during evacuation are expressed through the weighted contribution these indicators receive within the exposure component rather than through a separate vulnerability dimension.

The two SEIFA indices require separate treatment despite both being socioeconomic measures: SEIFA IER is placed under vulnerability on the basis that economic deprivation reduces a household's capacity to invest in flood protection, maintain insurance or fund post-event recovery (Aerts et al. 2018; Luu et al. 2020), which corresponds to the IPCC characterisation of vulnerability as the conditions that worsen impact, whereas SEIFA IEO is placed under exposure because it characterises the socioeconomic profile of the population present in the flood zone and reflects differential access to risk information and preparedness resources rather than the conditions governing the severity of impact on that population. Both indices combine multiple census dimensions within a single composite ranking, which avoids the multicollinearity that would arise from incorporating the constituent variables as separate indicators (ABS 2023). Rainfall, finally, is not used directly as an indicator, since the flood depth and flood spread derived from the HEC-RAS simulation represent the physical output of the precipitation event and therefore the flood itself rather than its meteorological precursor, a distinction that becomes consequential when comparing studies in which rainfall appears under hazard (Joo et al. 2024) and those in which it appears under exposure (Saha et al. 2024).

2.2.2 Data processing at the SA1 neighbourhood scale

All indicators were computed at the Statistical Area Level 1 using data sources described in Supplement Table S1. This is to capture intra-urban disparities that would be obscured at coarser geographic scales. Continuous raster-derived indicators, including flood depth, elevation, slope, curvature and drainage density, are summarised as the mean value within each SA1 polygon using zonal statistics. Area-based indicators, including vegetation density, transport density, building density and flood spread, are expressed as the percentage of the SA1 area occupied by the relevant feature class. Population-based indicators are expressed either as a proportion of the total resident population of the SA1 or as the number of residents

per unit area, as appropriate to the variable in question, while SEIFA IER and IEO scores were used as raw national ranking values, each placing the SA1 in relation to all other SA1 neighbourhoods across Australia.

2.3 Phase 2: Data Collection and Processing

2.3.1 Flood simulation using HEC-RAS

The October 2022 Maribyrnong flood event, the third largest recorded for the catchment (Water 2023), was selected as the basis for the hazard simulation because it generated the most complete sensor record available, with continuous hourly rainfall, river level and flow velocity readings from Melbourne Water gauging stations spanning 13 October 2022 to 15 October 2022 which were the dates of the flood event, and because its magnitude is sufficient to produce inundation across the full extent of the study area rather than confining flood waters to the immediate river corridor. These sensor records, together with the 10 m DEM as the terrain surface, were used as inputs to a HEC-RAS hydraulic simulation to generate the flood inundation dataset.

Since the definition of flooding adopted here requires water to be present under extraordinary rather than ordinary circumstances (Bakhtiari et al. 2023), the permanent watercourse network represents a condition of the landscape rather than a flood event and its pixels were therefore removed from the inundation raster using the mapped watercourse shapefile (DTP 2024) before indicator extraction. This is to have the resulting flood depth and spread values reflect only the inundation occurring beyond the normal channel boundaries. Flood spread was defined as the proportion of each SA1 area covered by water at depths of 0.15 m or greater, a threshold at which pedestrian and vehicular movement becomes significantly impaired (SES 2022), while mean flood depth was computed as the average inundation depth across all inundated cells within each SA1.

In the absence of a validated post-event aerial or satellite inundation record for the October 2022 event, the simulated inundation extent was assessed against the 1% Annual Exceedance Probability flood extent shapefile provided by Melbourne Water and Jacobs (2024) (Water and Jacobs 2024), which represents the design flood boundary for the catchment, a comparison that reflects differences in event magnitude as well as model performance and whose results should therefore be interpreted accordingly. The comparison between the two flood extents in Fig. 3 indicates the HEC-RAS output is consistent with the validation shapefile for the main area affected by the flood.

2.4 Phase 3: Normalisation and Weighting

2.4.1 Normalisation

Before the indicators could be combined into component indices, each was normalised to a common scale using the min-max method, in which values are rescaled to the interval [0, 1]. This prevents distortion between the difference in measurement units which would consequently affect the relative contribution of indicators to the final index. Because the indicators do not share the same directional relationship with flood risk, two forms of the normalisation were applied: for indicators where higher values correspond to

higher risk, Eq. 1 was applied directly, whereas for indicators where lower values correspond to higher risk, the result of Eq. 1 was subtracted from 1 so that the rescaled value increases with risk across all indicators consistently before aggregation.

$$Normalised(x) = \frac{x - \min(x)}{\max(x) - \min(x)}$$

1

Flood depth, flood spread, drainage density, population density, building density, dependent population and population with health condition received positive normalisation, while slope, elevation, curvature, Economic Resources, transport density, vegetation density and Education-Occupation Status received negative normalisation, consistent with the directional relationships set out in Table 1.

2.4.2 Weighting

The three risk components required different weighting approaches because the nature of the data and the measurement scale of the indicators differ fundamentally between them. The hazard indicators are all direct physical measurements of the same flood event, quantifying the depth, spatial extent and channel network associated with the October 2022 inundation, and it is not meaningful to weight one physical expression of the flood against another using a regression-based method since all three are manifestations of the same underlying process. Equal weighting was therefore applied to the hazard component, with the hazard index computed as the unweighted mean of the three normalised hazard indicators as shown in Eq. 2.

$$Hazard\ Index = \frac{Depth_N + Spread_N + Drainage\ Density_N}{3}$$

2

For the vulnerability and exposure components, the indicators span socioeconomic, demographic and physical variables whose relative importance varies across the study area in ways that a single global weight cannot capture, since a weight derived from the full study area averages out the local conditions that determine how strongly each indicator relates to flood risk in any given neighbourhood. Multiscale Geographically Weighted Regression (MGWR) recovers this spatial variation by estimating a separate regression coefficient for each indicator at every SA1 neighbourhood, where those coefficients, once normalised, become the neighbourhood-specific weights. Unlike standard GWR, which imposes a single bandwidth on all variables, MGWR allows each indicator to operate at its own spatial scale. This enables elevation, which varies gradually across the catchment, to be estimated at a broader scale while population density, which varies locally, is estimated at a finer one (Fotheringham et al. 2017; Wu et al. 2024).

Two MGWR models were estimated separately, one for the vulnerability indicators and one for the exposure indicators, each with flood spread as the dependent variable and the normalised indicators of

the respective component as the independent variables. The response variable in both models is flood spread, which measures the proportion of each SA1 area covered by inundation and provides a spatially complete record of where the October 2022 flood waters reached across all 412 neighbourhoods. Flood depth was considered as an alternative but was not used because the HEC-RAS simulation produced reliable depth estimates only within the immediate floodplain corridor of the Maribyrnong River, whereas the inundation extent covered the full study area and could therefore support coefficient estimation at every SA1.

The coefficients returned by MGWR carry both magnitude and sign, where a positive coefficient indicates that higher values of the indicator are associated with greater flood spread and a negative coefficient indicates the inverse relationship. Since the normalisation procedure in Section 2.4.1 has already standardised all indicators where higher normalised values correspond to higher risk regardless of the original direction of the relationship, the sign of the MGWR coefficient reflects the raw data relationship rather than the risk direction and cannot be used directly as a weight without producing negative index values in neighbourhoods where the coefficient is negative. The absolute value of each coefficient was therefore taken before weight calculation to have the weight represent the magnitude of the indicator's spatial association with the inundation footprint rather than its direction, which has already been accounted for in normalisation.

Indicator weights for each SA1 neighbourhood were derived by dividing each indicator's absolute coefficient by the sum of the absolute coefficients of all indicators within the same component for that neighbourhood. Eq. 3 demonstrates how the weights for each component sum to unity within every SA1 and are therefore directly comparable across neighbourhoods.

$$w_{i,SA1} = \frac{|C_{i,SA1}|}{\sum |C_{j,SA1}|}$$

3

where $w_{i,SA1}$ is the weight of indicator i in a given SA1 neighbourhood, $C_{i,SA1}$ is the MGWR coefficient for that indicator in that neighbourhood, and the summation in the denominator runs across all indicators j belonging to the same risk component. The weighted indicator value for each SA1 was then obtained by multiplying the normalised indicator value by its neighbourhood-specific weight, as in Eq. 4.

$$Weighted\ Indicator = Normalised(x) * w_{i,SA1}$$

4

The outcome of this phase is that each of the 412 SA1 neighbourhoods carries a unique set of indicator weights for the vulnerability and exposure components, derived from the local spatial relationship between each indicator and the inundation footprint rather than from a globally assumed level of importance. The weighted indicator values produced by Eq. 4 encode both the magnitude of the indicator at that location and the degree to which it is associated with the presence of flood water in that

neighbourhood's particular physical and socioeconomic context. These weighted values are the inputs to the component index calculation in Phase 4.

2.5 Phase 4: Index Calculation

2.5.1 Component indices

The weighted indicator values produced in Phase 3 were aggregated into component indices by taking the mean of the weighted indicators within each component for each SA1 neighbourhood, a procedure that treats each indicator's weighted value as its proportional contribution to the component rather than as an absolute score. Consequently, the index reflects the combined local influence of all indicators rather than being dominated by any single one. The general form of the component index calculation is given in Eq. 5, where i denotes the SA1 neighbourhood and n is the number of indicators in the component, which differs across the three components because the indicator set was defined by the physical and socioeconomic characteristics of the catchment rather than by a requirement for equal representation.

$$Component\ Index_i = \frac{\sum Weighted\ Indicators_i}{n_i}$$

5

For the hazard component, whose three indicators were weighted by the equal weighting procedure of Phase 3, Eq. 5 takes the form of Eq. 6.

$$H_i = \frac{\sum Weighted\ Hazard\ Indicators_i}{3}$$

6

For the vulnerability component, whose six indicators were weighted by the MGWR-derived neighbourhood-specific weights of Phase 3, the same form yields Eq. 7.

$$V_i = \frac{\sum Weighted\ Vulnerability\ Indicators_i}{6}$$

7

For the exposure component, whose five indicators were similarly weighted by MGWR-derived coefficients, Eq. 8 is used.

$$E_i = \frac{\sum Weighted\ Exposure\ Indicators_i}{5}$$

8

To return the three component index values from each of the 412 SA1 neighbourhoods, one for each dimension of risk, whose magnitudes reflect both the local presence of the underlying phenomena and the degree to which those phenomena are spatially associated with the inundation footprint in that neighbourhood's particular context. Before these three values were combined, each component index was normalised to the interval [0, 1] using the min-max procedure of Eq. 1. This prevents differences in the range and distribution of the component indices from distorting their relative contribution to the final risk score.

2.5.2 Flood risk index

The question of how to weight the three components relative to one another in the final risk index is one that the existing literature does not resolve, with some studies applying equal weights across components (Armenakis et al. 2017) and others using expert judgement of the kind the present framework is designed to avoid (Berrezel et al. 2026). Since the MGWR-derived weights already capture the spatially varying contribution of each indicator within its component, applying a further differential weighting between hazard, vulnerability and exposure would require either an additional model or a subjective assignment of component importance, neither of which is consistent with the objective approach adopted here, and the three normalised component indices were therefore combined as an unweighted mean to produce the flood risk index, as in Eq. 9, where FRI_i is the flood risk index for SA1 neighbourhood i , and H_N , V_N and E_N are the normalised hazard, vulnerability and exposure indices respectively, each constrained to [0, 1] where the resulting flood risk index lies within the same interval, with values approaching 1 indicating the highest flood risk.

$$FRI_i = \frac{H_N + V_N + E_N}{3}$$

9

The index does not represent the probability of a flood occurring but rather the combined magnitude of the physical hazard, the presence of exposed people and property, and the conditions that worsen their situation, assessed locally at the neighbourhood scale, and it is in that sense that the spatial distribution of its values, examined in Section 3, can be used to identify which neighbourhoods face the greatest risk and through which component that risk is largely expressed. Eq. 10 is the flood risk index equation for the full study where FRI = Flood Risk Index, i = Number of observations/SA1 neighbourhoods, H = Hazard, V = Vulnerability, E = Exposure, w = Weight, N = Normalised indicator.

$$FRI_{i=412} = \frac{1}{3} \left(\sum_{j=1}^5 w_E N_E + \sum_{j=1}^6 w_V N_V + \frac{1}{3} \sum_{j=1}^3 N_H \right)_i$$

10

3. Results

3.1 Indicator data

The processed indicator values across the 412 SA1 neighbourhoods are summarised in Table 2. For most indicators the mean value lies considerably closer to the minimum than to the maximum, reflecting the skewed distributions that characterise urban catchment data where extreme values occur in a small number of neighbourhoods, typically those located on or immediately adjacent to the floodplain, while the majority of the study area returns moderate to low values. The SEIFA-derived indicators and building density are exceptions to this pattern: SEIFA scores are national rankings rather than physical measurements and are distributed by construction across a wide range, while building density reflects the predominantly urban land cover of the catchment where the absence of buildings is the exception rather than the rule. The inverse relationship between vegetated area, building area and road area per SA1 is consistent with the land cover composition expected in an urbanised catchment, where the progressive replacement of permeable surfaces with impervious cover reduces the capacity of the landscape to absorb and retain floodwater (Miranda et al. 2023).

Table 2

Descriptive statistics for the processed indicator values across the 412 SA1 neighbourhoods.

Component	Indicator	Minimum	Mean	Maximum
Hazard	Flood depth (m)	0.02	0.86	7.87
	Flood spread (% per SA1)	0	39	100
	Drainage density (km^{-1})	0	0.65	3.11
Vulnerability	Slope ($^{\circ}$)	0.02	2.16	11.75
	Elevation (m)	1.62	38.49	105.49
	Curvature (m^{-1})	-0.18	0.003	0.23
	Economic Resources (SEIFA IER)	443	983	1221
	Vegetation density (% per SA1)	0.03	7.57	26.29
Exposure	Transport density (% per SA1)	0.99	23.85	51.48
	Population density (people per m^2)	0	0.42	7.95
	Building density (% per SA1)	0.57	28.57	55.17
	Dependent population (% per SA1)	0	36.77	100
	Population with health condition (% per SA1)	0	5.98	47.16
	Education–Occupation Status (SEIFA IEO)	712	1053	1201

3.2 MGWR model performance

The MGWR models for both vulnerability and exposure returned improved goodness of fit relative to the standard GWR models estimated on the same variables and response variable, confirming that the multiscale specification, in which each indicator is allowed to operate at its own spatial scale rather than sharing a single bandwidth with all other variables, better captures the spatial structure of the relationships between the indicators and the inundation footprint across the study area. Table 3 presents the fit statistics for both components.

Table 3 Comparison of goodness of fit statistics of GWR and MGWR model performance for vulnerability and exposure components.

Statistic	Vulnerability		Exposure	
	GWR	MGWR	GWR	MGWR
R^2	0.8972	0.8982	0.6551	0.7382
Adjusted R^2	0.8498	0.8676	0.5485	0.6885
AICc	876.37	847.30	1288.76	1144.50
Optimal bandwidth (k-nearest neighbours)	57	69	57	69

Notes: R^2 = coefficient of determination; Adjusted R^2 = adjusted coefficient of determination; AICc = corrected Akaike Information Criterion.

The vulnerability model returned an Adjusted R^2 of 86.76% under MGWR compared with 84.98% under GWR, with the AICc falling from 876.37 to 847.30, and the stronger fit relative to the exposure model can be attributed to the predominantly physical nature of the vulnerability indicators, elevation, slope, curvature, vegetation density and transport density, whose spatial variation is gradual and systematic across the catchment and therefore more readily captured by a regression-based spatial model. The exposure model returned a larger improvement under MGWR, with Adjusted R^2 increasing from 54.85% to 68.85% and AICc falling from 1288.76 to 1144.50, a gain that reflects the benefit of allowing population-based indicators, which vary at finer spatial scales than the physical terrain, to operate at bandwidths appropriate to their own scale of variation rather than being constrained by the single bandwidth imposed in standard GWR.

3.3 Spatial distribution of risk components

The spatial distribution of the three component indices, shown in Fig. 4, reveals distinct patterns for each component that reflect the different physical and socioeconomic processes governing their variation across the catchment.

The vulnerability index is highest in a broadly horizontal belt across the lower centre and lower right of the study area, a pattern that corresponds closely to the distribution of low elevation terrain and, to a lesser extent, to areas of lower Economic Resources scores in the southern neighbourhoods, where the combination of flat topography, limited vegetation cover, low slope and reduced economic capacity produces index values that no single indicator would generate alone. The north-western and north-central areas return the lowest vulnerability values, consistent with their higher elevation and greater transport density, and the strong spatial correspondence between high vulnerability and high flood depth, despite depth not entering the vulnerability calculation directly, reflects the physical logic of the catchment in which the same low-lying terrain that prolongs inundation also concentrates the conditions that worsen its impact.

The exposure index displays considerably greater spatial heterogeneity than vulnerability, with the highest values clustered in the central-western area of the study area and a secondary cluster in the eastern centre, while the south-eastern industrial area and the north-western boundary return the lowest values, the former because it carries significant human activity but few permanent residents and low building density relative to its SA1 area, the latter because it is more distant from the river corridor and has lower population concentration. The absence of a clear spatial gradient in the exposure map, in contrast to the more systematic pattern of the vulnerability map, reflects the dominance of population-based indicators whose distribution is governed by residential land use patterns rather than by the physical gradient of the terrain.

The hazard index is, as expected, highest along the Maribyrnong River corridor and in the downstream areas where the river converges with Steele Creek, with values decreasing with distance from the waterway network. A patch of moderately high hazard is present on the southern edge which can be attributed to moderate to high spread and drainage density values, despite depth being relatively low there. The Flemington Racecourse neighbourhood returns a high hazard index as a consequence of its position on the floodplain downstream of the Maribyrnong township, a finding that becomes significant when its exposure and vulnerability values are considered alongside it in Section 3.4.

3.4 Flood risk index

The flood risk index map, shown in Fig. 5 returns the highest values in the central area of the study area along the Maribyrnong River corridor and in the south-eastern section where the Yarra and Maribyrnong Rivers converge, with risk decreasing with distance from the main waterway network. Kensington records the highest mean flood risk index across all SA2 neighbourhoods at 0.46, a value driven principally by its vulnerability component, which at 0.81 is the highest in the study area and reflects its position in a low elevation area directly downstream of the Maribyrnong floodplain, combined with moderately high building

and population density. West Footscray has the second highest mean risk at approximately 0.41, notable because the neighbourhood is not situated adjacent to any waterway, and its high risk derives from elevated vulnerability and exposure rather than from the hazard component, which at 0.06 is among the lowest in the study area.

The violin plot in Fig. 6 illustrates the range and distribution of risk values within each SA2 region. Kensington has the widest range among all SA2 regions, reflecting the diversity of conditions within its 22 SA1 neighbourhoods, which span both the high-risk floodplain neighbourhoods along the river and the lower-risk areas on its periphery. West Footscray has a narrow range and consistently high values, indicating that the elevated risk is a characteristic of the area as a whole rather than being concentrated in a subset of its neighbourhoods. Flemington Racecourse, by contrast, returns a high hazard value but a low overall risk score because its exposure index of 0.00 reflects the absence of resident population and the near-absence of buildings on the racecourse itself, a result that illustrates the importance of considering all three components simultaneously rather than treating hazard as a proxy for risk.

The bar graph in Fig. 7 confirms that vulnerability and exposure are the dominant drivers of flood risk across the SA2 neighbourhoods, with hazard consistently returning the smallest component values and the two socioeconomic components producing the peaks and troughs that determine the final risk index. This pattern holds even in areas historically associated with flooding: Flemington and Footscray, both situated in low elevation downstream areas with high hazard exposure, owe their elevated risk scores more to the density of people and property and the physical conditions that worsen impact than to the magnitude of the flood waters themselves, a finding that has direct implications for where resilience investment is most likely to reduce risk most efficiently.

Two areas emerge from the analysis as previously unidentified high-risk zones that have not historically been associated with flooding. West Footscray and Tottenham, and the southern and south-eastern neighbourhoods of Braybrook (SES 2021), return high risk values despite their distance from the main waterway network, a pattern attributable to their downstream position relative to the Maribyrnong floodplain, their low elevation and flat terrain, and their moderate drainage density, conditions that are sufficient to generate elevated vulnerability and exposure index values even in the absence of a direct flood history.

3.5 Spatial distribution of indicator weights

The MGWR-derived weights for the three most influential indicators, population density, vegetation density and building density, are shown in the supplementary file (Supplementary S14) and illustrate the degree to which the relative importance of each indicator varies from one neighbourhood to the next in ways that a single global weight cannot capture.

Population density weights are highest in the central and western neighbourhoods where residential density is greatest and where the spatial covariance between population concentration and inundation extent is strongest, and lowest in the airport precinct and industrial areas where population data are unreliable, consistent with the negative local R^2 values identified in Section 3.2. Vegetation density returns the widest spatial variation in weights of all the indicators, with the highest weights concentrated in the small neighbourhoods on the south-western boundary and a central neighbourhood where vegetation cover is both present and spatially associated with inundation extent, while the large open neighbourhoods on the north-eastern and south-eastern boundaries return low to moderate weights despite having higher vegetation cover, a divergence attributable to the difference in neighbourhood size and the corresponding dilution of the local spatial relationship in larger SA1 areas. Building density weights follow a pattern broadly consistent with the distribution of urban land cover across the catchment, with higher weights in the densely built central and western areas and lower weights in the peri-urban north, though the relationship is modulated by the local inundation footprint where building-dense neighbourhoods outside the flood extent return lower weights than building-dense neighbourhoods within it.

Taken together, the weight maps demonstrate that the framework produces a robust local assessment in which the contribution of each indicator to the component index reflects the conditions of each neighbourhood rather than a study-wide assumption, and that the spatial heterogeneity of those contributions is sufficiently pronounced to produce materially different risk rankings from those that a uniform-weighting approach would generate.

4. Discussion

4.1 Key findings and mitigation implications

The spatial distribution of flood risk across the 412 SA1 neighbourhoods of the Maribyrnong River catchment is neither uniform nor fully explained by proximity to the waterway network, and it is this disparity that the framework is designed to reveal. The highest risk values are concentrated in Kensington, whose mean flood risk index of 0.46 reflects a combination of low elevation, high population density and limited Economic Resources that produces vulnerability and exposure conditions severe enough to sustain high risk even in periods between flood events, and in the downstream areas of Maribyrnong and Avondale Heights where the physical hazard is greatest. The lowest values occur in the north-western and north-central areas of the catchment where higher elevation, greater transport density and lower residential density reduce all three components simultaneously. The spatial gradient in risk broadly follows the physical gradient of the terrain but is modulated by the socioeconomic characteristics of the population in ways that terrain alone cannot predict.

Population density, building density and vegetation density are the indicators with the greatest average influence on the exposure and vulnerability components respectively, and their dominance is consistent across the catchment in the sense that they return the highest mean coefficient values in the MGWR models, but the weight maps reveal that this dominance is not spatially uniform. Population density

weights are highest in the central and western residential areas where population concentration and inundation extent spatially coincide most strongly, and lowest in the airport precinct and industrial zones where the spatial relationship between population and flood water is weakest, a variation that a global weighting approach would obscure by assigning the same importance to population density regardless of whether the neighbourhood in question is a dense residential area or a largely uninhabited industrial zone. Vegetation density displays the widest weight variation of all indicators, reflecting the heterogeneous land cover of a catchment that encompasses dense urban fabric, open parkland and suburban gardens within a relatively small geographic area, and its spatial variation in weight is the clearest illustration in this study of why a locally adaptive approach produces a materially different assessment from a uniform one.

The neighbourhoods facing the greatest flood risk are those in Kensington, West Footscray and Tottenham, and the southern and south-eastern areas of Braybrook, and the last two of these represent a finding that conventional hazard-proximity assessments would not have produced since neither area has a flood history and neither is situated adjacent to a waterway. Their elevated risk derives from a convergence of downstream position, flat low-lying terrain, moderate drainage density and high exposure values that together produce index scores comparable to neighbourhoods on the floodplain itself, and it is precisely because their risk is driven by vulnerability and exposure rather than by hazard that it would remain invisible to a hazard-focused assessment. For local government and emergency services, the implication is that resource allocation and community engagement strategies based on flood history or proximity to waterways will systematically underserve these neighbourhoods, and that a component-level assessment of the kind developed here is necessary to identify where the greatest need lies and through which dimension of risk it is principally expressed. Neighbourhoods with high dependent population and population with health condition concentrations, such as those in Avondale Heights, require particular attention in evacuation planning given the lower mobility of these groups during flood events (Ajtai et al. 2023; Rana et al. 2023), while neighbourhoods with low Economic Resources scores, concentrated in the southern areas of the catchment, face compounded vulnerability from both their physical exposure and their limited capacity to invest in protection or recovery (Aerts et al. 2018). Prioritising those neighbourhoods is essential for mitigating risk as the population residing in those areas would take longer time recovering from the flooding event if they are affected first.

The relative contribution of the three components to the final risk index varies considerably across the catchment, and the contrast between Flemington Racecourse and the adjacent Flemington SA2 region provides the clearest illustration of why all three components must be considered simultaneously. Flemington Racecourse returns the highest hazard index value in the study area, consistent with its documented flood history (Water 2023), but its exposure index of 0.00 reflects the near-complete absence of resident population and permanent buildings on the racecourse itself, so that a high physical hazard produces a low overall risk score in a location that is, in the conventional sense, the most flood-affected site in the catchment. Flemington, by contrast, returns a low hazard index but a high vulnerability and exposure index from the density of its residential population and built environment, producing a higher overall risk score than the racecourse despite its lower direct flood exposure, a result that confirms the finding across multiple studies that the socioeconomic dimensions of risk, rather than the physical

magnitude of the flood, determine the actual consequences of an event for a community (Armenakis et al. 2017; Feng et al. 2023). Vulnerability being the highest contribution to risk can be attributed to how it is influenced by geographical and urban factors. Conversely, hazard is geographically limited to waterway proximity while further away neighbourhoods will have its hazard solely determined by rainfall.

4.2 Methodological contributions

The main theoretical contribution of this study is that it demonstrates a path between the two positions that dominate the flood risk weighting literature: the subjective but locally sensitive approach of the Analytical Hierarchy Process, whose weights reflect expert judgement and apply uniformly across the study area, and the objective but spatially insensitive approach of equal weighting, which treats all indicators as equally important regardless of local context. The MGWR-based weighting procedure adopted here produces weights that are both objective, in the sense that they are derived from the spatial structure of the data rather than from expert opinion, and locally varying, in the sense that each SA1 neighbourhood receives a unique set of weights that reflects its particular relationship with the inundation footprint. The Adjusted R^2 values of 86.76% for vulnerability and 68.85% for exposure confirm that the spatially varying coefficients provide a reliable empirical basis for weighting in this catchment, substantially outperforming the conceptual PCA-GWR framework proposed by Chen (2021), whose highest Adjusted R^2 was 0.49, and the improvement of MGWR over standard GWR in both models confirms that the multiscale specification is warranted given the different spatial scales at which physical and socioeconomic indicators operate across an urbanised catchment.

The technical contribution of the framework lies in the combination of three design choices that individually appear in the literature but have not previously been combined in a single assessment. The use of IPCC definitions for component classification provides a standardised basis for indicator allocation that addresses the inconsistency documented by Feng et al. (2023) and demonstrated in this study's own review of the literature, where rainfall, vegetation and population health indicators appear under different components across studies. The application of neighbourhood-specific weights rather than global weights provides a level of spatial granularity that AHP-based frameworks, which assign one weight per indicator across the entire study area, cannot achieve, and the use of the additive plus model for combining components, in which hazard, vulnerability and exposure each contribute one third of the final index, ensures that areas with low hazard but high vulnerability and exposure are not systematically under ranked, as they would be in multiplicative frameworks where a near-zero hazard value suppresses the final score regardless of the other components (Ahmed et al. 2022). This framework provides insights on the contribution of vulnerability, hazard and exposure, and their individual indicators have on the risk index combined with the local level of analysis thoroughly assesses which indicators and components influences risk the most and the neighbourhoods that are facing the highest level of risk.

4.3 Limitations and future research

The most consequential limitation of the framework as implemented here is that the MGWR models use flood spread as the response variable rather than flood risk itself. This response variable was used to have the indicator weights reflect each indicator's spatial covariance with the inundation footprint rather than

with a measure of risk that already incorporates all three components. This is an inherent constraint of the framework's sequential structure, in which the weights must be derived before the risk index can be calculated, and it means that the weights capture the local relevance of each indicator to the physical presence of floodwater rather than to the full risk experienced by the neighbourhood's population. An iterative approach, in which a preliminary equal-weighted risk index serves as the response variable in a second round of MGWR estimation, would address this limitation and represents the most direct extension of the present framework, though it would require careful attention to the circularity introduced by using a risk index as both an input and an output of the weighting procedure.

Several further limitations warrant acknowledgement. Similar to the MGWR model, flood depth was used as the response variable rather than risk itself for the OLS tests. The three population indicators (population density, dependent, with health condition) were not statistically significant in the OLS test. This can be attributed to all three indicators being population-based and the response variable being flood depth rather than flood risk where no relationship would be expected. The Moran's I result expectedly returned the relationship between exposure indicators and flood depth all occurred by chance based on its z-score, p-value and Moran's I. Although the same occurred with vulnerability, the values are close to being considered dispersed as the z-score is -1.49 and p-value is 0.14, despite it indicating statistical insignificance. This can be attributed to how the vulnerability indicators are predominantly from the physical environment which influences flood behaviour such as elevation, slope and curvature. Exposure is conversely people and property indicators which is what is affected. The HEC-RAS output does not have comparison data for the full study area. This means comparing and validating the generated flood extent will be difficult as the shapefile used only accounts for the lower study area. HEC-RAS did not generate an RMSE or error file to display which areas would cause uncertainties in accuracy.

The 84 exposure neighbourhoods returning negative local R^2 values, concentrated in the Melbourne Airport precinct and the eastern study area boundary, indicate areas where the model's explanatory power is limited and where the derived weights should be treated with caution; the underlying cause, which is the unreliability of resident population data in areas dominated by non-residential activity, points to a broader limitation of census-based exposure indicators in mixed-use urban environments. The temporal inconsistency between the 2021 Census and SEIFA data and the October 2022 flood simulation introduces a minor source of uncertainty in the exposure and vulnerability weights, though the broad socioeconomic structure of the catchment is unlikely to have changed materially across this period. The equal weighting of the three components in the final risk index, while consistent with the objective approach of the framework, remains an assumption that future work could address through a component-level MGWR model or through sensitivity analysis of alternative component weightings.

Two directions for future research follow directly from the limitations identified here. The first is the incorporation of resilience as a fourth dimension of the risk equation, in which the framework's risk index serves as the input to a resilience-adjusted score of the form proposed by Schwarz et al. (2023), where $\text{Risk} = (\text{Hazard} \times \text{Exposure} \times \text{Vulnerability}) - \text{Resilience}$, factoring in the mitigating effect of existing flood protection infrastructure, emergency response capacity and community preparedness in the final assessment rather than being treated as a background condition. The second is the expansion of the

indicator set to include built environment variables beyond road and building density, drawing on the framework proposed by Wang et al. (2025), whose Australian nationwide assessment of social vulnerability identifies access to public facilities, utilities and greenspace, diversity in land use, housing tenure and commercial density as dimensions of vulnerability that SEIFA indices do not capture but that are directly relevant to flood preparedness and recovery capacity. The inclusion of these indicators would address the limitation that SEIFA, while providing a reliable composite measure of socioeconomic status, lacks the built environment dimensions that determine whether a community has the physical infrastructure to respond effectively to a flood event.

5. Conclusion

This study has developed a spatially adaptive flood risk framework for the Maribyrnong River catchment that derives indicator weights from the spatial structure of the data rather than from expert judgement, producing a neighbourhood-scale assessment in which the contribution of each indicator to the component index varies across the 412 SA1 neighbourhoods in ways that reflect local physical and socioeconomic conditions rather than a single assumed level of importance. The framework demonstrates that vulnerability and exposure exert greater influence on flood risk than hazard alone across the study area, that population density, building density and vegetation density are the dominant risk drivers, and that the spatial heterogeneity of indicator weights is sufficiently pronounced to identify high-risk neighbourhoods, most notably in West Footscray, Tottenham and the southern areas of Braybrook, that a hazard-proximity or uniform-weighting assessment would not have surfaced, since their risk derives from the convergence of downstream position, flat terrain and concentrated exposure rather than from a direct flood history.

The replicability of the methodology across other urbanised catchments is its most practically significant property, since the combination of IPCC-standardised component definitions, MGWR-derived local weights and a publicly available sensor and census data infrastructure means that the framework can be applied wherever equivalent data exist without requiring expert elicitation or subjective judgement at any stage of the process. For policymakers and emergency services working in catchments where flood risk is unevenly distributed and where the communities most exposed are not necessarily those closest to the water, the framework provides a basis for identifying which neighbourhoods to prioritise, which dimension of risk to address first, and how the relative importance of individual indicators shifts from one part of the catchment to another, equipping resilience planning with the local specificity that uniform assessment methods, however carefully constructed, cannot provide.

Declarations

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This study does not involve human or animal subjects; therefore, ethical approval is not applicable.

Consent for publication:

Not applicable.

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The authors declare no competing interests.

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Author Contribution

Nicholas Lee: Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing, Data curation, Validation and Formal analysis

Chayn Sun: Conceptualization, Methodology, Validation, Writing - Reviewing & Editing, Supervision, Project Administration

Monica Wachowicz: Conceptualization, Methodology, Validation, Writing - Reviewing & Editing, Supervision

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Data Availability

The Census, SEIFA, VicMap terrain, gauging-station records, and land-cover datasets, and building footprint data analysed in this study are publicly available from the sources cited in the manuscript. The

flood extent shapefile and council 3D building data are not publicly available due to licensing restrictions imposed by the data providers (Melbourne Water and Jacobs, and the City of Maribyrnong and Moonee Valley councils) but are available from the corresponding author on reasonable request and with the permission of those providers.

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Figures

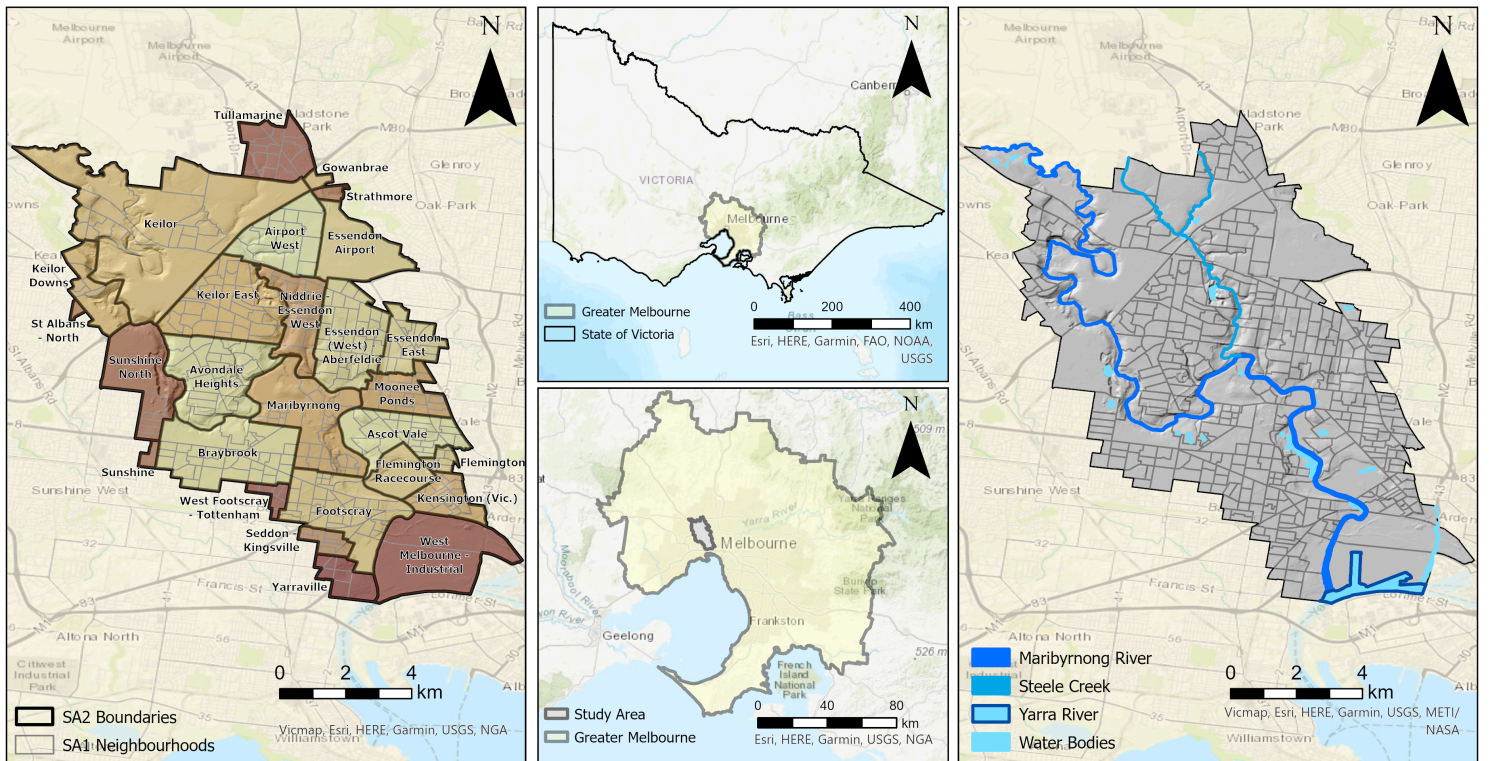


Figure 1

Geographic representation of the study area map. The left panel shows the spatial distribution of SA2 and SA1 boundaries, while the right panel displays the waterways of the study area.

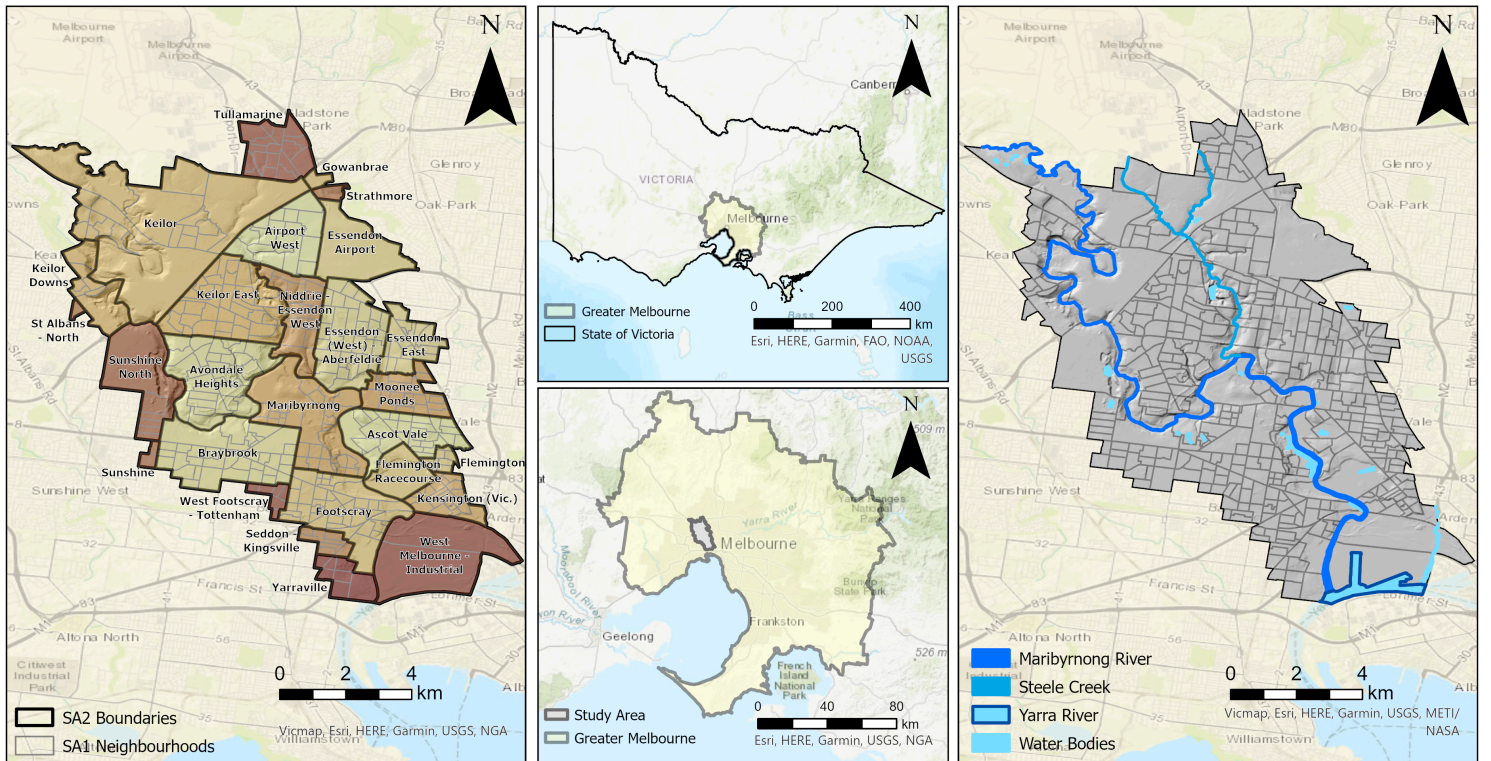


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Geographic representation of the study area map. The left panel shows the spatial distribution of SA2 and SA1 boundaries, while the right panel displays the waterways of the study area.

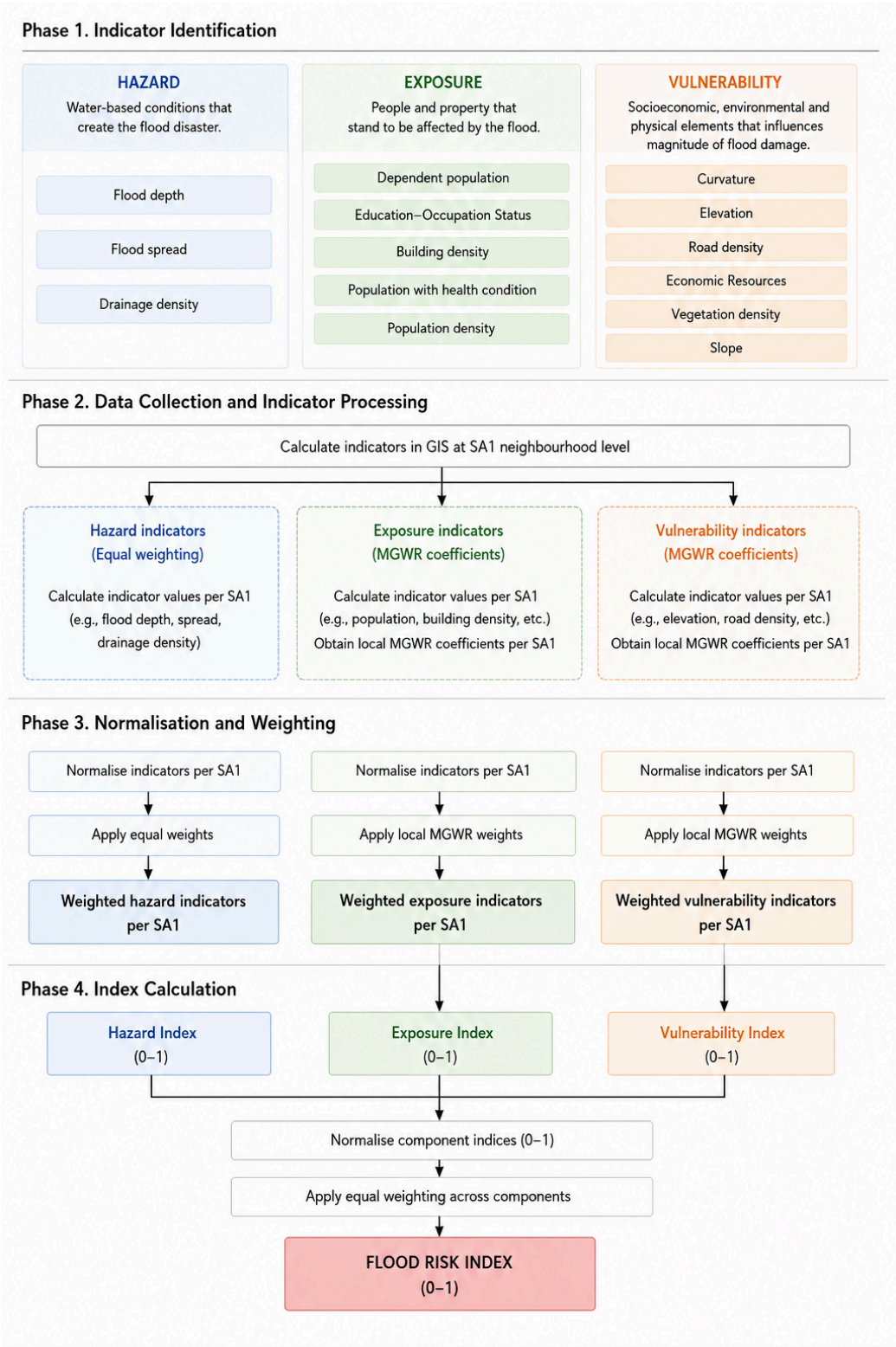


Figure 2

Methodological framework for flood risk assessment, showing the hazard, exposure, and vulnerability components and the four-phase workflow used to calculate the Flood Risk Index.

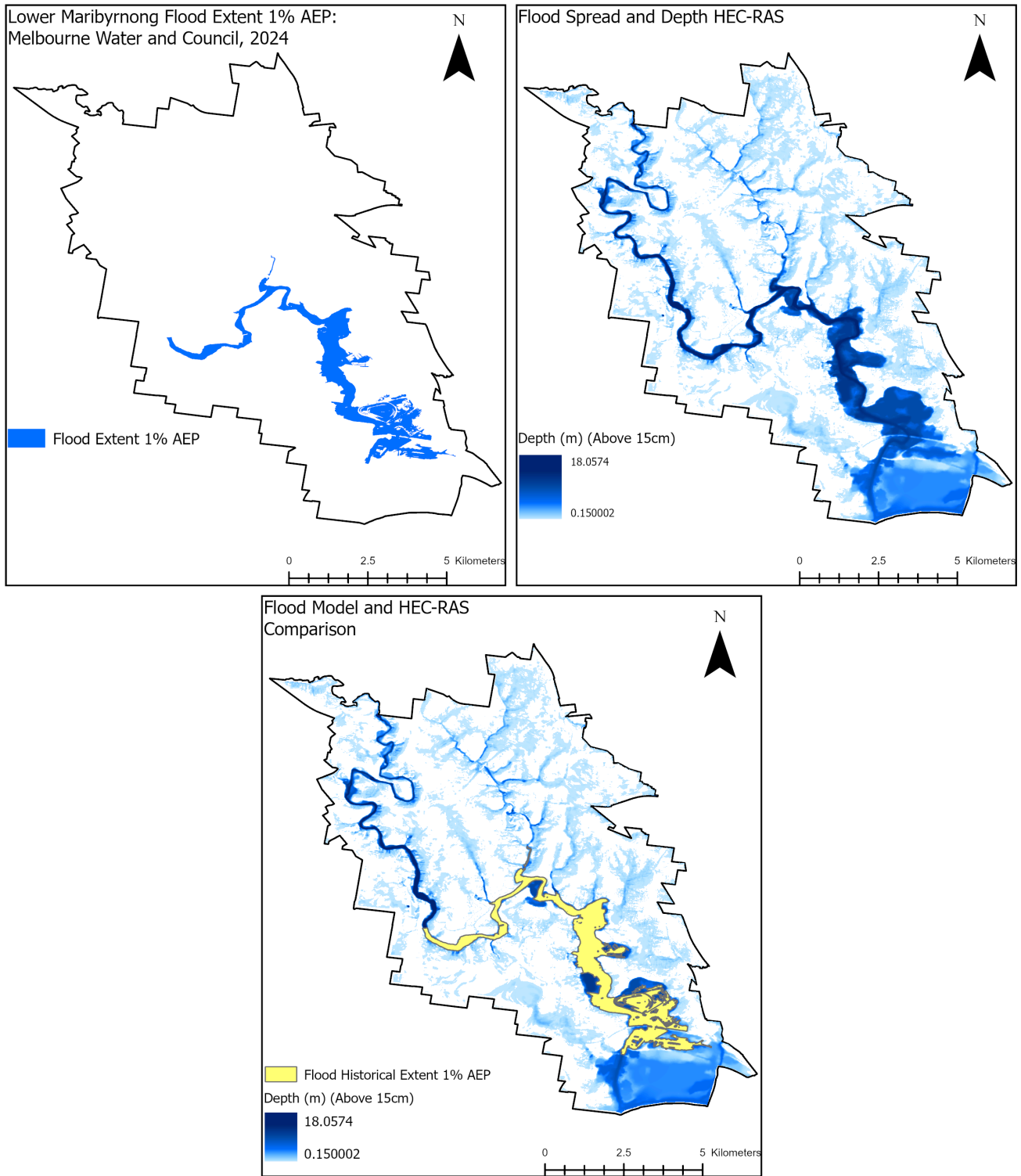


Figure 3

Comparison map of a flood extent shapefile from an external entity (top-left), the flood extent and depth from HEC-RAS (top-right) and the shapefile overlaid with the HEC-RAS output (bottom).

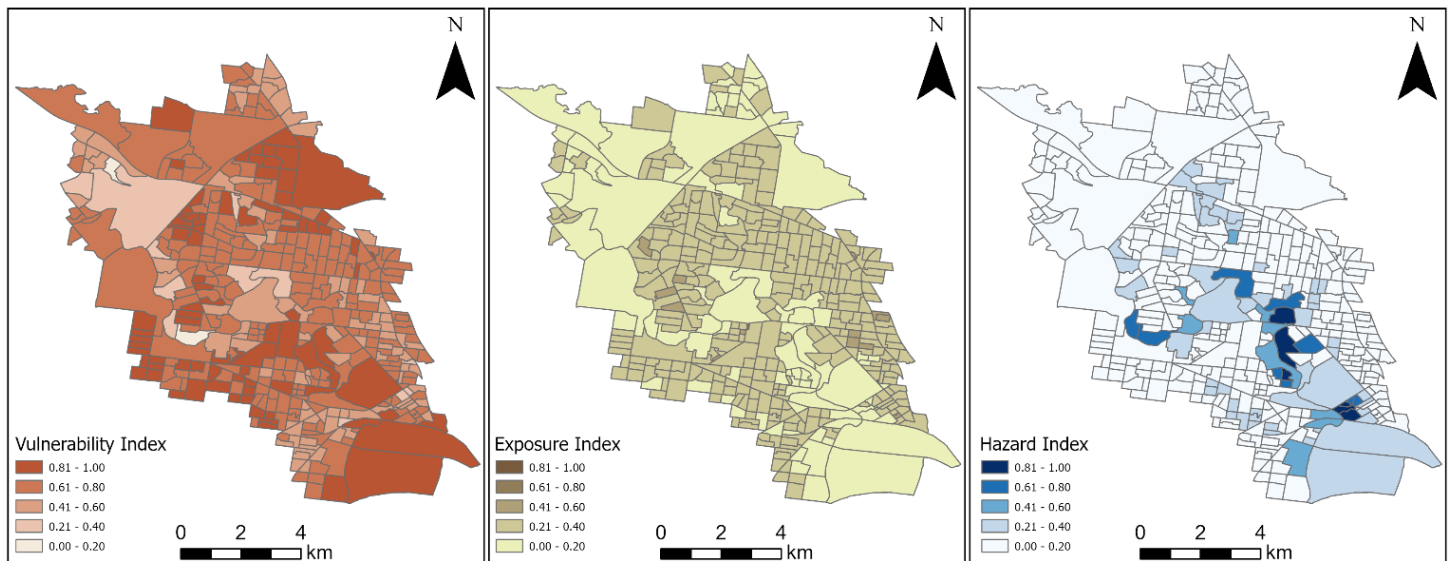


Figure 4

Spatial neighbourhood distribution of the hazard, exposure, and vulnerability components across the study area.

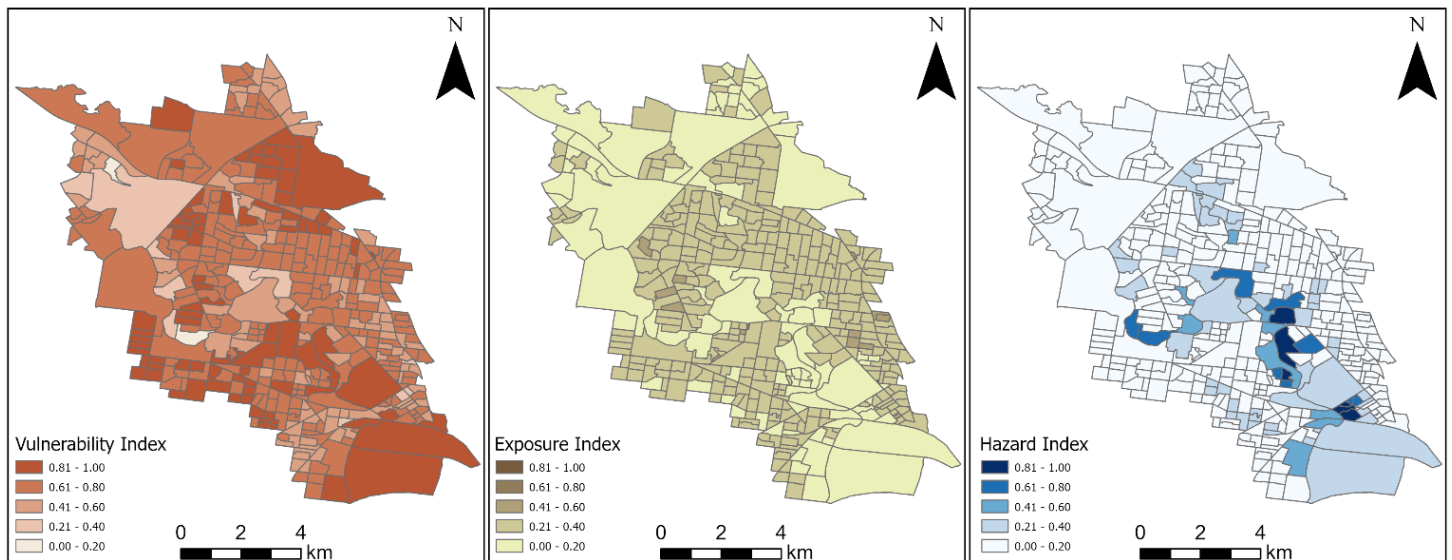


Figure 4

Spatial neighbourhood distribution of the hazard, exposure, and vulnerability components across the study area.

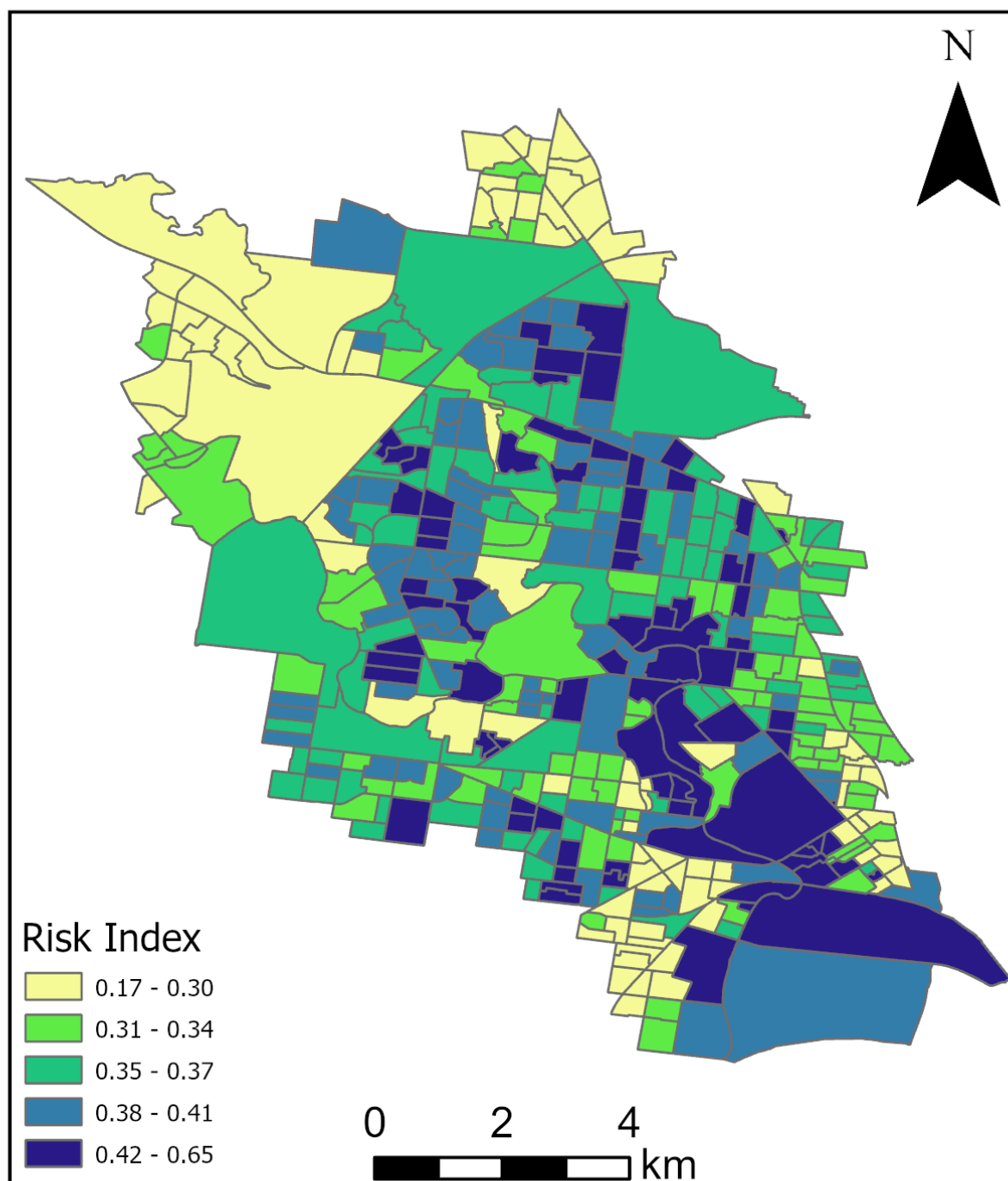


Figure 5

Spatial distribution of the Flood Risk Index across SA1 neighbourhoods in the study area.

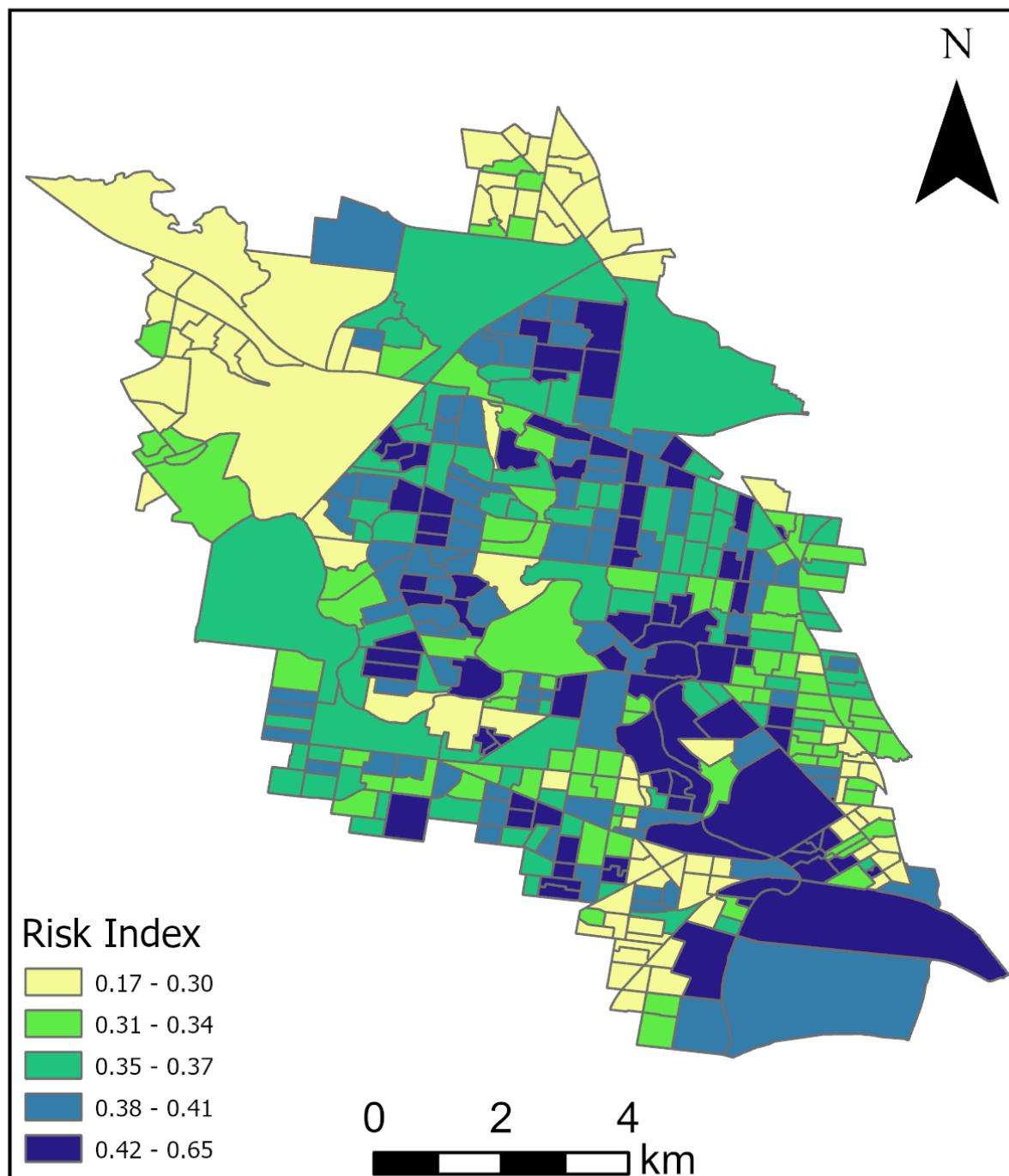


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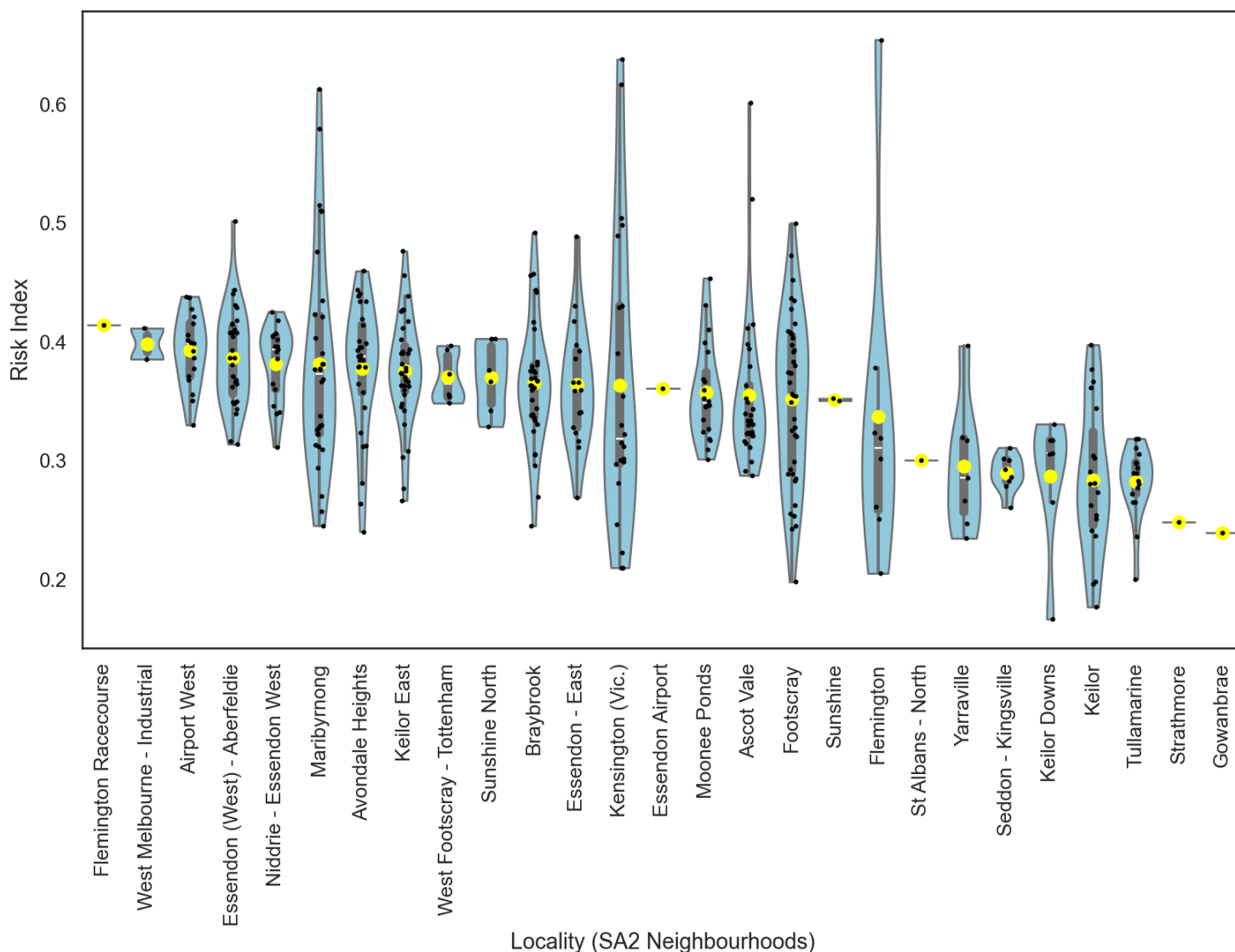


Figure 6

Violin plot of risk distribution per locality. Yellow dot indicates mean risk value. White line indicates the median value. Individual dots represent the risk value of each neighbourhood within the locality.

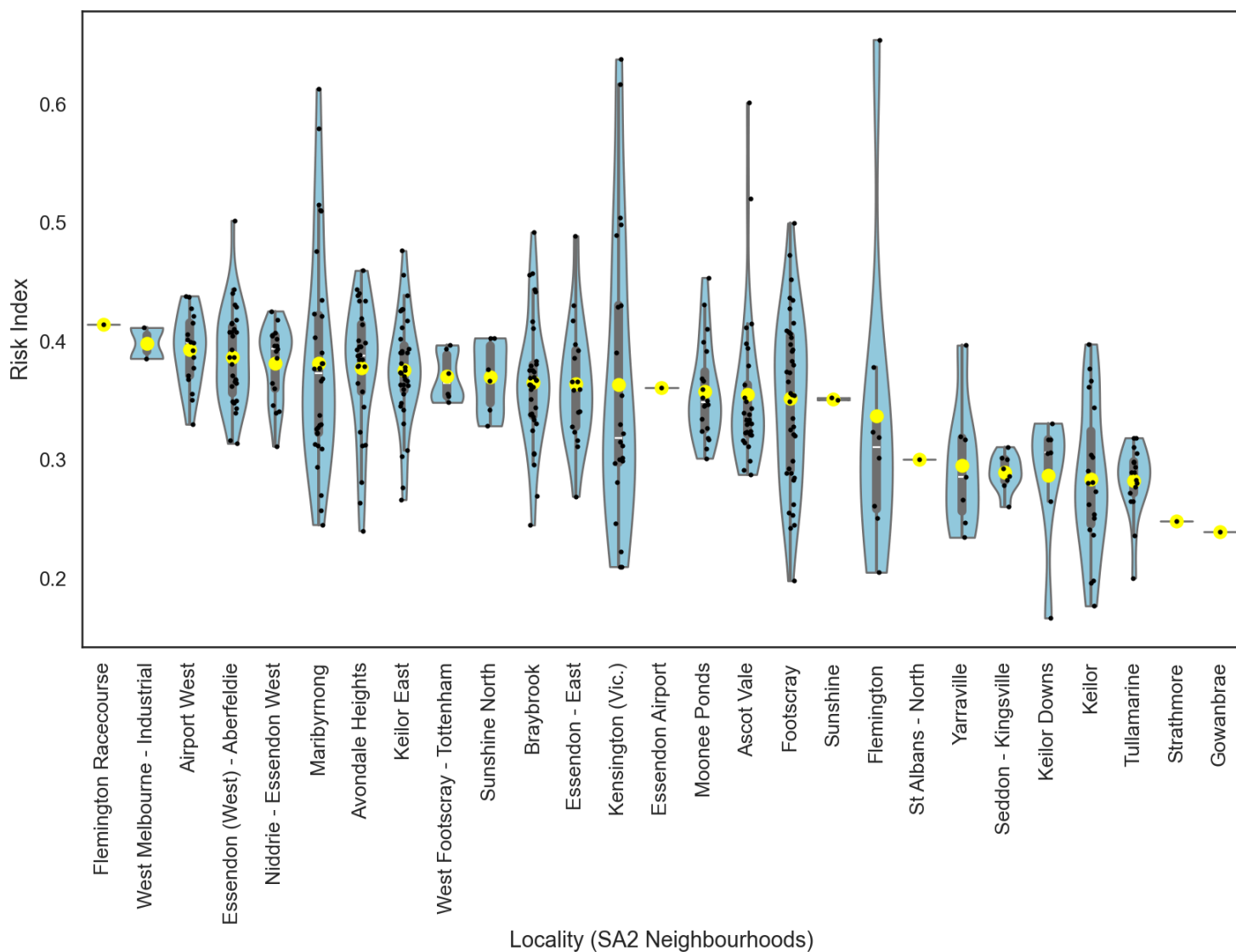


Figure 6

Violin plot of risk distribution per locality. Yellow dot indicates mean risk value. White line indicates the median value. Individual dots represent the risk value of each neighbourhood within the locality.

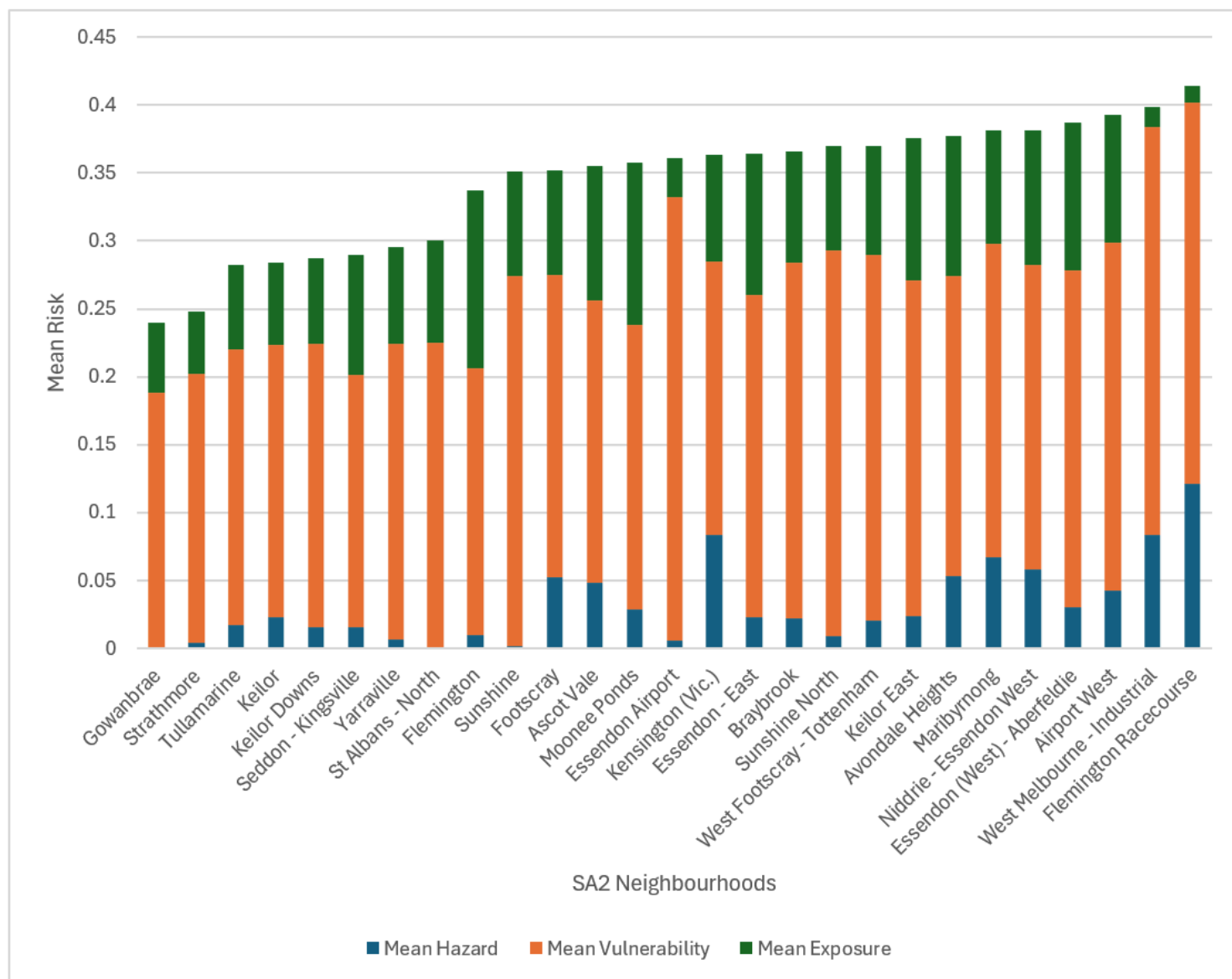


Figure 7

Mean risk components across SA2 neighbourhoods.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [SupplementaryFileFloodriskassessmentattheneighbourhoodlevelusingSpatiallyAdaptiveWeighting.pdf](#)